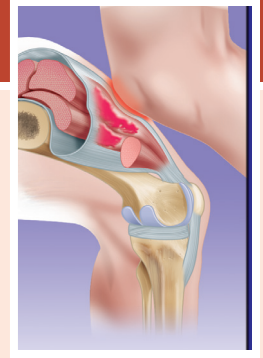


The Athletic Trainer

Chad Starkey, Shannon David



The name *athletic trainer* (AT) implies that ATs coach and train athletes to improve baseline performance. This title and the euphemism “trainer” create ambiguity regarding the AT’s knowledge, skills, and abilities. ATs are multiskilled health care clinical professionals who deliver services in cooperation with and under the direction of physicians to provide optimal patient care. The AT’s education and scope of practice encompass the areas of injury prevention and risk management, clinical examination and diagnosis, emergency care, therapeutic intervention (i.e., therapeutic modalities and rehabilitation), and health care administration (Box 30.1).

Although they are most visible when working with high school, collegiate, and professional sports teams, ATs are employed in a variety of settings, including physicians’ offices, clinics, hospitals, performing arts, and the military.¹ ATs tend to work with a highly motivated, physically active population.

ATs can obtain third-party reimbursement in many settings. However, an advantage of ATs employed by high schools and intercollegiate athletics is the provision of capitated health care services. In the case of high schools located in poor and medically underserved areas, the presence of an AT is a cost-effective approach to the health care of these students. They often serve as the first point of medical contact for the community at large.

The knowledge, skills, and abilities that an AT possesses are found in other health care professions; however, the specific combination of skills is unique to AT. Because ATs have their roots in athletics, with a reasonable sense of urgency to return patients to competition, ATs have developed a philosophy of aggressive intervention that benefits both “athletic” and “nonathletic” patients. The AT skillset is applied according to the AT philosophy of an aggressive, yet safe, return to activity. This approach is based on the diagnostic principles and patient’s intervention strategies and goals established by the World Health Organization’s International Classification of Functioning, Disability, and Health.² The role of AT in the overall health care community is often misunderstood because of the overlap of skills with other professions, portrayals in movies and television (see “The Knute Rockne Story” and “The Water Boy”), the lack of understanding of the AT philosophy of care, and the misleading name, “athletic training” (K. K. Knight, C. Starkey, and D. Fandel, unpublished manuscript, 2009).

BRIEF HISTORY OF ATHLETIC TRAINING

The origin of athletic training can be traced back to the ancient Olympics, when paleotribes (loosely translated to “boy rubber”) assisted athletes with their health care. In the United States the roots of athletic training emerged in the early 1900s, when individuals began to assist coaches and physicians in caring for the medical needs of athletes. In 1950, the National Athletic Trainers’ Association was formed to help guide the practice of athletic training, which at the time was primarily limited to collegiate and professional teams.³

Similar to physical therapy and occupational therapy, the first athletic training academic programs have their roots in physical education. During the past 25 years, athletic training has progressed through an academic major to a formal degree and is currently transitioning to a professional (entry-level) master’s degree to be eligible to practice as an AT. The focus of classroom and clinical education was once singularly focused on athletes. However, advancements in medicine and health care have extended the age of people participating in athletics and other forms of strenuous physical activity. Improved health care has also decreased the number of conditions that can disqualify a person from competitive athletics. In response, ATs have increased their knowledge of the unique challenges faced by people who have underlying medical or physical limitations throughout the life span.

Although the “traditional” athletic population remains a central theme of education, ATs have evolved to develop expertise in the care of a broad segment of the physically active population. This expanded educational base has changed the employment patterns of ATs, extending well beyond high school, collegiate, and professional team sports medicine venues to include industrial, military, the National Aeronautics and Space Administration (NASA), physician’s practices, among others.

In just over half a century, ATs have evolved from the locker room to become a health care provider recognized by the American Medical Association. Contemporary ATs incorporate current evidence and best practices to treat a physically active patient base to ensure physical readiness to return to their desired level of function after injury (Box 30.2).^{4–12}

EDUCATION

ATs must graduate from a professional bachelor’s or master’s degree program accredited by the Commission on Accreditation

Abstract

The profession of athletic training has evolved from “locker room helpers” to a full-fledged health profession that requires a professional degree (soon moving to the master’s entry-level), certification, and a state license to practice. Working under the direction of a physician (MD or DO), athletic trainers (ATs) nowadays are multitasked providers who deliver health care services to athletes, industrial workers, the military, and other physically active people. The AT’s skillset includes emergent and emergency care, clinical diagnosis, interventions such as therapeutic exercise, therapeutic modalities, and manual skills required to return the individual to full activity, as well as administrative and psychosocial training. In settings such as high schools and colleges, ATs are the point of first medical contact. Regardless of the work setting, ATs are uniquely positioned to extend the physician’s services.

Keywords

education
credentialing
role delineation
skills
abilities

BOX 30.1 Core Educational Competencies

- Evidence-based practice
- Interprofessional education and collaborative practice
- Quality improvement
- Health care informatics
- Professionalism
- Patient-centered care

BOX 30.2 Athletic Training's Research Contributions

The emergence of academic degree programs has resulted in the development of scholars who contribute to the sports medicine knowledge base. Athletic trainers are at the forefront of research regarding the prevention and diagnosis of conditions affecting athletes and others engaged in strenuous physical activity. Athletic training researchers were among the first to question the long-term consequences of athletic-related concussions and to question the efficacy of the clinical examination techniques used to identify these conditions.^{4–12} Other researchers have added to the evidence base for orthopaedic diagnostic techniques, therapeutic interventions, and immediate care of musculoskeletal injuries.

Based on a strong, multidisciplinary evidence base, The National Athletic Trainers' Association has developed position statements regarding topics such as prevention of heat illness and sudden death, the management of concussions, athletes with cervical spine injuries, and athletes with type 1 diabetes, disordered eating, and asthma. Athletic trainers also have worked with other medical organizations on consensus statements regarding heat acclimation, the prehospital care of athletes with a spine injury, and athletes with sickle cell trait. For more information, see <http://www.nata.org/membership/membership-benefits/athletic-training-publications>.

Clinicians have the obligation to remain up-to-date regarding the current standard of care and best practices (via position statements) described in the prior paragraph. In addition, athletic trainers are required to obtain 10 hours of evidence-based practice continuing education every 2 years.

of Athletic Training Education (CAATE). By 2022, professional education will occur at the master's degree level. The educational content of an AT program is defined by the CAATE accreditation Standards,¹³ whereas the Role Delineation Study¹⁴ defines base entry-level practice. **Box 30.1** presents the content area required in the professional preparation of ATs. **Table 30.1** presents the prerequisite and foundational knowledge required in educational programs.

Professional coursework spans a range of musculoskeletal, neurologic, and metabolic conditions seen across the life span and focuses on the continuum of integrated prevention, care, and return to activity. The minimum program requirement is 2 years of classroom and clinical education with a proportion being in an interprofessional environment. A sampling of categories of professional education include:

- Developing emergency action plans (EAPs)
- Injury prevention including preparticipation examinations
- Clinical diagnosis and emergent care and appropriate referral
- Developing a plan of care for patients with conditions involving multiple systems
- Diagnostic testing (imaging, blood work, urinalysis, electrocardiography) to facilitate clinical diagnosis, referral, and plan of care

TABLE 30.1 Required Prerequisite and Foundational Knowledge

Prerequisite Coursework	Foundational Knowledge
Biology	Statistics and research design
Chemistry	Epidemiology
Physics	Pathophysiology
Psychology	Biomechanics and pathomechanics
Human anatomy	Exercise physiology
Human physiology	Nutrition
	Human anatomy
	Pharmacology
	Public health
	Health care delivery and payor systems

Note:

Prerequisite coursework is required prior to entering the professional program.

Foundational knowledge may be prerequisite coursework or incorporated into the professional program.

- Passive, active, and manual intervention to restore function (e.g., therapeutic modalities, therapeutic exercise, manual therapies)
- Knowledge of pharmacologic agents
- Concussion management and education
- Behavioral health conditions
- Durable medical equipment, orthotics, bracing, protective padding
- Mitigation of risk for long-term health conditions
- Environmental conditions
- Drug use/abuse education

These domains are tied together through the use of evidence-based practice. Many ATs are world-class scholars who have emerged as leaders in the diagnosis and management of concussion,¹⁵ heat illness,¹⁶ cervical spine trauma,¹⁷ and therapeutic interventions.¹² ATs also consume research produced by physicians, physical therapists, and other professions who address the needs of persons who are physically active.

More than 70% of ATs possess advanced degrees, including accredited postprofessional programs (master's and doctoral degree programs and residencies).¹⁸ Many currently credentialed ATs are dual credentialed, most often in conjunction with the fields of physical therapy, physician's assistant, and/or strength and conditioning/performance enhancement.

Regulation

Graduation from an accredited professional program is a requisite to sit for the Board of Certification, Inc. (BOC) examination as an entry-level AT. The "ATC" designation indicates that a person has passed the BOC examination, which serves as the common examination for individual states that issue the practice credential, typically "LAT" or "AT."¹⁹ Although most states have licensure for professional practice, some states regulate practice via Registration or State Certification (**Table 30.2**). As of 2017 California is the only state that does not have any sort of state regulation of AT practice. Although the language of these practice acts is broad, a commonality is that ATs work collaboratively with and/or under a physician's direction.

TABLE 30.2 State Regulation of Athletic Trainers

Type	Description
Licensure	Licensure restricts practice to persons who have meet the licensing board's requirements. The practice act describes the athletic trainer's scope of practice. Unlicensed persons are prohibited from practicing athletic training.
Certification	Similar to state licensure, persons must meet minimum educational requirements and pass a state examination (the Board of Certification, Inc., examination is often recognized for this purpose). However, state certification only provides title protection; it does not limit uncertified persons from practicing.
Registration	Registration may or may not have educational or examination requirements. By registering with the state, title protection is granted.
Exemption	Exemption excludes a person from the standards of other licensed professions (e.g., physician assistant, physical therapy, or nursing).

Modified from Ray RR, Konin J. *Management Strategies in Athletic Training*. Champaign, IL: Human Kinetics Publishers; 2011.

PHYSICIAN/ATHLETIC TRAINER
WORKING RELATIONSHIP

Regardless of the workplace setting, ATs function to extend the physician's services, serving as the physician's eyes, hands, and ears. The AT frequently is the point of first contact for injured/ill individuals. This role is unique because ATs often perform the first examination of an injury, usually minutes after its onset. This role triages the referral process, expediting those patients who need immediate medical assistance and preventing the physician's and patient's time loss in the event of needless referrals. In addition, this relationship has been shown to improve patient care and save the physician time.²⁰ Physicians have also reported better quality of life when they incorporate ATs in their practice.²¹

ATHLETIC TRAINER SCOPE OF PRACTICE

A unique aspect of athletic training is that, in many instances, ATs follow their patient throughout the continuum of care, from preinjury (prevention) through the diagnostic and intervention stages to the return-to-activity decision. The actual scope of practice within each state may differ from the description provided in this section. Physicians who direct AT practice should consult the state's AT practice act for applicable regulations. Specific scope of practice questions should be directed to the state practice board (<http://www.bocalc.org/athletic-trainers#state-regulation>). The following is a brief description of skills an AT practices on a daily basis.

Injury Prevention and Health Promotion

The basis of injury prevention and risk management is ensuring the individual's physical readiness to participate in strenuous activity, ensuring a safe playing/work environment, and developing and implementing EAPs. Another form of injury prevention

is ensuring the safe return to activity after an injury has been sustained. Lastly, ATs spend significant time educating patients before, during, and after injury and illness.

Using both written health questionnaire and physical examination, the preparticipation physical examination identifies a person's physical readiness to engage in selected activities. During a routine preparticipation physical examination, the patient's and patient's family medical history are analyzed, and a review of systems and regional examinations are performed. In addition, baseline testing such as concussion testing and strength and range of motion (ROM) assessments may be conducted.

ATs work with physicians, administrators, and attorneys to develop EAPs. The EAP describes the standard of care and the procedures to follow in the event of foreseeable emergent situations (e.g., cardiac arrest, cervical spine injury), inclement weather (e.g., heat or lightning), or other possible venue-specific contingencies.

Patient education is the most encompassing method of injury prevention and risk management. When working with high school-aged patients (or younger), patient education also includes the athlete's parent(s) or legal guardian. The AT is often the primary source for information regarding concussions, heat illness, sickle cell trait, and nutritional and hydration needs. This role also extends to bridging the gap between the patient, the patient's family, and the physician regarding the potential outcomes of surgery or other interventions for an injury or illness (and, likewise, the possible consequences of not following the physician's advice). Because ATs tend to see their patients on a frequent (sometimes daily) basis, they serve as a valuable resource for answering questions that may arise during the patient's treatment.

Clinical Diagnosis and Immediate Care

The immediate (on-field) examination first rules out life- or limb-threatening conditions, fractures, or dislocations. The on-field examination ultimately culminates in the decision about how to remove the athlete from the playing field (e.g., assisted or unassisted) and whether the condition requires that the patient be immediately transported to a hospital for emergency care.

The clinical examination relies on obtaining a medical history and performing a functional assessment, inspection, palpation, and assessment of joint and muscle function to form a differential diagnosis. Joint-specific stress tests, selective tissue tests, and, when applicable, neurologic and vascular tests are used to rule in or rule out various pathologies, resulting in a working clinical diagnosis. The AT triages the patient and determines if a referral is indicated. Once a diagnosis has been established, the AT may consult with the physician to determine the appropriate course of care (Fig. 30.1).

Immediate and Emergent Care

In cases in which the findings of the acute (i.e., immediately following the injury) clinical examination and diagnosis indicate that the patient requires emergent care and/or direct transportation to a medical facility, the AT leads (or provides assistance during) the process.^{22,23} The scheme of the on-field examination follows the principle of ruling out life-threatening conditions



Fig. 30.1 Athletic trainers possess a range of skills that include clinical examination and diagnosis, immediate and emergent care, therapeutic interventions, and administrative skills.



Fig. 30.2 Athletic trainers work clinically and in the field.

and conditions that jeopardize the integrity of the extremity and then performing a more finite musculoskeletal examination. In equipment-intensive sports such as football and hockey, the player's protective equipment may hinder the examination process.

Even before arriving at the scene, the AT begins the triage (i.e., the primary assessment of airway, breathing, and circulation, shock, severe bleeding, and spinal injury). If the athlete is experiencing cardiorespiratory compromise, life support is begun. If this initial screen is negative, the athlete's level of consciousness is ascertained. If the athlete is unconscious (or if a cervical spine injury is indicated), the AT stabilizes the cervical spine and leads the process of immobilizing the patient on a spine board. The AT is often the only person on site to make the return-to-play (RTP) determination, which is particularly important in the case of concussion management. ATs are well educated in the diagnosis and management of concussion, along with RTP criteria.

The on-field orthopaedic examination is more rote than a clinical examination. Although its underlying purpose is to determine if assistance is needed to transport the athlete to the sideline or directly to a medical facility (and, if so, the transportation method required), it is also the best opportunity to identify joint stability before a muscle spasm sets in. The focus is on the immediate history of the injury (primarily the mechanism), inspection and palpation for gross deformity, and determination of active ROM. Although exceptions occur, in general only uniplanar stress tests are performed on the field; selective tissue tests are reserved for a more controlled examination environment.

ATs are well educated regarding transportation techniques, ranging from spine boarding to manually assisted transport. When appropriate, the AT will apply an appropriate protective device and/or fit the patient with crutches to ensure safe ambulation.

Therapeutic Interventions

Therapeutic intervention includes therapeutic modalities, manual therapy, and therapeutic exercise to return the patient to the desired level of function. In a more athletic population, the primary emphasis is on reducing participation restrictions. However, the athletic training philosophy tends to yield

interventions that are more aggressive than those used with the general population. Although the protocols used are aggressive, the goal is to return the patient to activity safely and in the shortest time possible. Identification of the relative and absolute contraindications and precautions to intervention techniques is embedded in the patient examination.

Some therapeutic modalities, such as ice, moist heat, electrical stimulation, and therapeutic ultrasound, are passive devices used to regulate the physiologic response to trauma. Other modalities such as traction or compression are used to cause physical changes in the tissues. Typically, therapeutic modalities are used to allow active exercise or manual therapy to be performed.

Manual therapies include massage, myofascial release, augmented soft tissue mobilization, and joint mobilization (Fig. 30.2). These therapies are augmented by passive, active-assisted, active, and resisted ROM exercises. Once strength and ROM are restored, a functional progression relative to the body part that involves appropriate strengthening, proprioceptive activities, and activity-specific skills is implemented.

Prescription, over-the-counter, and herbal supplements are commonly used by athletes and other physically active persons. ATs understand the influence of pharmacologic approaches in the management of acute and chronic injury and disease and recognize the potential ramifications for patient care.

Psychosocial Strategies and Referral

Although the focus of athletic health care often focuses on physical injury and disease states, emphasis is growing on the mental and emotional components. Conditions such as drug use/abuse, disordered eating, and the female athlete triad have received substantial attention in professional journals and the lay press, and more recent attention has been placed on depression, anxiety, and stress. High school and collegiate student-athletes often turn to the AT as the point of first contact in seeking assistance. The AT does not presume to diagnose these conditions but rather intervenes on behalf of the individual to refer the patient to the proper provider for appropriate assistance.²⁴

During the rehabilitation process, ATs also use social support²⁵ and motivational techniques such as imagery, mental rehearsal, and goal setting to maximize the patient's progression through

the intervention program. These techniques often counter the sense of depression and isolation that some athletes experience after sustaining a significant injury.

Health Care Administration

Many athletic injuries require coordinated care involving physicians, surgeons, institutional and individual insurance providers, outpatient facilities, and billing departments. In many instances, the AT assumes the role of a case manager, acting as the liaison between the various medical personnel involved in the patient's care. Blending administration and education, the AT coordinates communication between the patient's parents, the coaching staff, and the media regarding the patient's disposition.

SUMMARY

ATs are state-regulated health care providers who extend the services of a physician. The AT's multidisciplinary educational background incorporates knowledge, skills, and abilities that include injury prevention, clinical diagnosis, emergency care, therapeutic intervention, and RTP decisions. The AT is an invaluable asset in the health care of athletes and other physically active persons.

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:

Schmidt JD, Register-Mihalik JK, Mihalik JP, et al. Identifying impairments after concussion: normative data versus individualized baselines. *Med Sci Sports Exerc*. 2012;44(9):1621–1628.

Level of Evidence:

I

Summary:

In the absence of individualized baseline tests, normative data may be used for neurocognitive testing, postural control, and graded symptom checklists when diagnosing a concussion.

Citation:

Herring SA, Cantu RC, Guskiewicz KM, et al. American College of Sports Medicine. Concussion (mild traumatic brain injury) and the team physician: a consensus statement—2011 update. *Med Sci Sports Exerc*. 2011;43(12):2412–2422.

Level of Evidence:

V

Summary:

This document is a revision of the 2006 Team Physician Statement regarding the diagnosis and management of concussions. This statement reinforces the recommendation of not returning to play the same day as the concussion, describes the role and limitations of neuropsychologic testing, and supports the need for “cognitive rest.”

Citation:

Hoch MC, Andreatta RD, Mullineaux DR, et al. Two-week joint mobilization intervention improves self-reported function, range of motion, and dynamic balance in those with chronic ankle instability. *J Orthop Res*. 2012;30(11):1798–1804.

Level of Evidence:

III

Summary:

In this study, the outcomes of Maitland grade III anterior/posterior joint mobilization treatments on ROM, star excursion balance test, and self-reported outcomes scales were compared in 12 subjects. The researchers concluded that the intervention resulted in improved scores for all measures.

Citation:

National Athletic Trainers' Association. *Athletic Training Education Competencies*. 5th ed. <http://www.nata.org/education/competencies>. Accessed July 19, 2013.

Level of Evidence:

V

Summary:

A multidisciplinary panel identified the knowledge, skills, and abilities that are required to be taught in an accredited professional (entry-level) athletic training program.

Citation:

Podlog L, Eklund RC. Return to sport after serious injury: a retrospective examination of motivation and psychological outcomes. *J Sport Rehabil*. 2005;14:20–34.

Level of Evidence:

II

Summary:

This article describes the effect of internal and external motivation factors in an athlete's return to participation after injury. The researchers found that intrinsic motivation led to more positive outcomes than did extrinsic motivation.

REFERENCES

1. National Athletic Trainers Association. Athletic training services: an overview of skills and services performed by certified athletic trainers; 2010. <http://www.nata.org/sites/default/files/GuideToAthleticTrainingServices.pdf>. Accessed July 19, 2013.
2. World Health Organization. International Classification of Functioning, Disability and Health (ICF). <http://www.who.int/classifications/icf/en/>. Accessed July 19, 2013.
3. Ebel RG. *Far Beyond the Shoe Box: Fifty Years of the National Athletic Trainers' Association*. New York: Forbes Custom Publishing; 1999.
4. Schmidt JD, Register-Mihalik JK, Mihalik JP, et al. Identifying impairments after concussion: normative data versus individualized baselines. *Med Sci Sports Exerc*. 2012;44(9):1621–1628.
5. Herring SA, Cantu RC, Guskiewicz KM, et al. American College of Sports Medicine: concussion (mild traumatic brain injury) and the team physician: a consensus statement—2011 update. *Med Sci Sports Exerc*. 2011;43(12):2412–2422.
6. Cantu RC, Guskiewicz K, Register-Mihalik JK. A retrospective clinical analysis of moderate to severe athletic concussions. *PM R*. 2010;2(12):1088–1093.
7. Guskiewicz KM, Valovich McLeod TC. Pediatric sports-related concussion. *PM R*. 2011;3(4):353–364.
8. Casa DJ, Armstrong LE, Kenny GP, et al. Exertional heat stroke: new concepts regarding cause and care. *Curr Sports Med Rep*. 2012;11(3):115–123.
9. DeMartini JK, Martschinske JL, Casa DJ, et al. Physical demands of National Collegiate Athletic Association Division I football players during preseason training in the heat. *J Strength Cond Res*. 2011;25(11):2935–2943.
10. Cosby NL, Koroch M, Grindstaff TL, et al. Immediate effects of anterior to posterior talocrural joint mobilizations following acute lateral ankle sprain. *J Man Manip Ther*. 2011;19(2):76–83.
11. Hoch MC, Andreatta RD, Mullineaux DR, et al. Two-week joint mobilization intervention improves self-reported function, range of motion, and dynamic balance in those with chronic ankle instability. *J Orthop Res*. 2012;30(11):1798–1804.
12. Merrick MA, McBrier NM. Progression of secondary injury after musculoskeletal trauma—a window of opportunity? *J Sport Rehabil*. 2010;19(4):380–388.
13. Commission on Accreditation of Athletic Training Education. Standards for the accreditation of professional athletic training programs; 2012. <https://caate.net/wp-content/uploads/2017/01/2012-Professional-Standards.pdf>. Accessed November 14, 2017.
14. Henderson J. *The 2015 Athletic Trainer Practice Analysis Study*. Omaha: Board of Certification; 2015. http://www.bocac.org/system/document_versions/versions/24/original/boc-pa7-content-outline-20170612.pdf?1497279231. Accessed November 14, 2017.
15. Guskiewicz K, et al. National Athletic Trainers' Association position statement: management of sport-related concussion. *J Athl Train*. 2004;39(3):280–297.
16. Binkley HM, Beckett J, Casa D, et al. National Athletic Trainers' Association position statement: exertional heat illness. *J Athl Train*. 2002;37(3):329–343.
17. Swarts E, et al. National Athletic Trainers' Association position statement: management of cervical spine-injured athlete. *J Athl Train*. 2009;44(3):306–331.
18. National Athletic Trainers' Association. Membership statistics. www.nata.org/sites/default/files/Profiles-of-Athletic-Trainers.pdf. Accessed September 26, 2013.
19. Board of Certification. State regulatory agencies. <http://www.bocac.org/athletic-trainers#state-regulation>. Accessed November 14, 2017.
20. Frogner BK, Westerman B, DiPietro L. The value of athletic trainers in ambulatory settings. *J Allied Health*. 2015;44(3):169–176.
21. Pecha FQ, Bahnmaier LA, Hasty MJ, et al. Physician satisfaction with residency-trained athletic trainers as physician extenders. *IJATT*. 2014;19(2):1–3.
22. Andersen JC, Courson R, Kleiner D, et al. National Athletic Trainers' Association: emergency planning in athletics. *J Athl Train*. 2002;37(1):99–104.
23. American College of Sports Medicine. Mass participation event management for the team physician: a consensus statement. *Med Sci Sports Exerc*. 2004;36(11):2004–2008.
24. Vaughan JL, King KA, Cottrell RR. Collegiate athletic trainers' confidence in helping female athletes with eating disorders. *J Athl Train*. 2004;39(1):71–76.
25. Barefield S, McCallister S. Social support in the athletic training room: athlete's expectations of staff and student athletic trainers. *J Athl Train*. 1997;32(4):333–338.